IoT-Enabled Smart bus route monitoring and revenue management system

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Project Proposal Report

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DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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2

ABSTRACT

Public bus transportation system in Sri Lanka plays a vital role in millions of people's daily lives. A massive portion of this current system is operated by private bus owners who focuses solely on profit which causes several problems for the government and the normal Sri Lankan citizen. One of the biggest problems for citizens is the lack of public bus availability during off-peak times (9PM to 5AM). Bus owners are only charged a fixed amount by the government for their permits regardless of the revenue generated by their buses. Which limits the revenue earned by the government. Additionally, some buses are known to deviate from their permitted road during off peak times which reduces service reliability. Already existing transportation management systems in Sri Lanka severely lack Real time location tracking, automated schedule generation and distance-based permit revenue calculation. This project aims to use digital technologies to improve efficiency, transparency and accountability in public bus operations. The proposed system introduces a GPS based revenue management system. A GPS device will be installed on buses to track their real time location and calculate total distance travelled. Permit revenue will be calculated through the data collected via the device. The system will also detect any route deviations and alert the governing bodies instantly while creating flexible timetables to ensure 24-hour service. A mobile app that serves bus owners and drivers is also developed in order to keep the entities updated on permit fees and other important information. The project will utilize GPS tracking modules, Cloud- based data bases and GIS systems to socially enhance the reliability and accessibility of public transformation while commercially creating a data driven regulatory model for sustainable transport management.

Keywords: Public Bus Route Tracking, Transport Management, Permit Revenue Calculation

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Table of Contents

ABSTRACT 3

ACKNOWLEDGEMENT 4

LIST OF TABLES 6

LIST OF FIGURES 7

LIST OF ABBREVIATIONS 8

LIST OF APPENDICES 9

1. INTRODUCTION 10

1.1 Background and Context 10

1.2 Literature Review 14

1.3 Research Gap 15

2. OBJECTIVE 17

2.1 Main Objective 17

2.2 Specific Objectives 17

3. METHODOLOGY 18

4. HIGH-LEVEL SYSTEM ARCHITECTURE 22

5. USER REQUIREMENTS 23

6. COMMERCIALIZATION PLAN 25

7. BUDGET AND JUSTIFICATION 27

8. WORK BREAKDOWN STRUCTURE (WBS) 28

9. GANTT CHART 30

10. REFERENCES LIST 31

11. APPENDICES 32

LIST OF TABLES

Table 1- Component & justification 20
Table 2- Platform & Justification.....20
Table 3- Tools & Justification.....20
Table 4- FR1 23
Table 5- FR2 23
Table 6- FR3 23
Table 7- FR4 23
Table 8- FR5 24
Table 9- FR6 24
Table 10- FR7 24
Table 11- FR8.....24
Table 12- Budget Estimation 27
Table 13- WBS.....29

LIST OF FIGURES

Figure 1 - Bus Schedule 11
Figure 2- Bus Schedule 12
Figure 3- High-level Architecture 22
Figure 4 - WBS chart 28
Figure 5 - Gantt Chart 30

LIST OF ABBREVIATIONS

SDG – Sustainable Development Goals

DB - Database

ETA – Estimated Time of Arrival

IS – Information System

ITS – intelligent transportation system

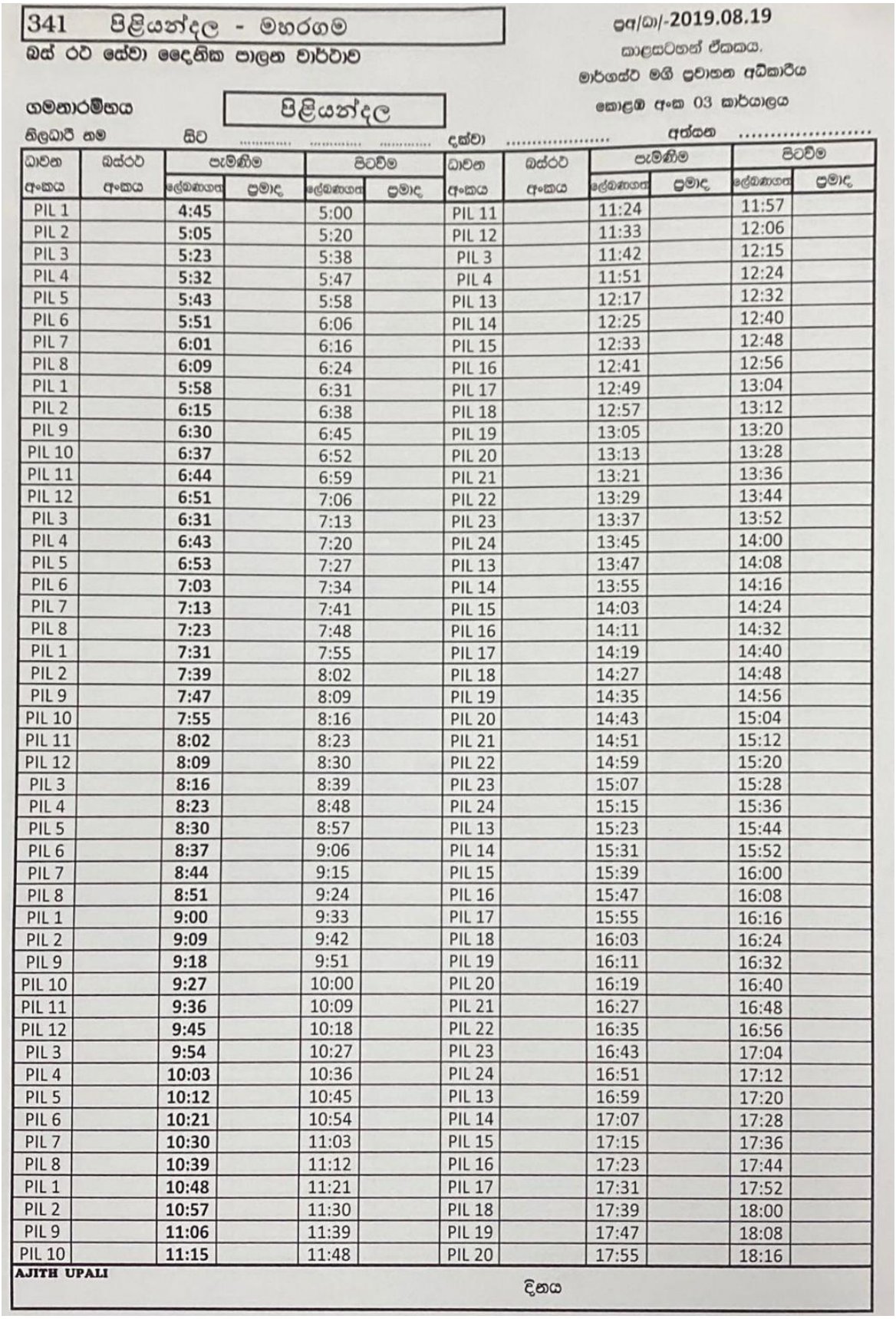
LIST OF APPENDICES

Appendix	Title	Page
Appendix A	Survey / Interview Instruments	32
Appendix B	Detailed System Architecture Diagram	32
Appendix C	Preliminary UI Mockups	32

1. INTRODUCTION

1.1 Background and Context

The backbone of any efficient urban mobility system is the public transportation. Millions of people, particularly students, workers and communities with low income, depend highly on affordable public transport here in Sri Lanka. A large share of this much needed service is operated by private bus owners with more than 21000 buses. Even though highly profited, bus owners only have to pay a very low permit fee (fixed annual and monthly fee) in order to keep operating, which reduces the income received by the government. With the disproportionately low income from private sector, funds available for transport infrastructure and regulations are highly reduced. This issue leads to one of the biggest problems in Sri Lanka's public transport, the unavailability during off peak times (9PM-6AM). Since private buses are profit oriented, bus owners are less likely to run during off peak times and with their low income, the government also face issues with operating SLTB buses to answer people's needs effectively.



341 පිළියන්දල - මහරගම

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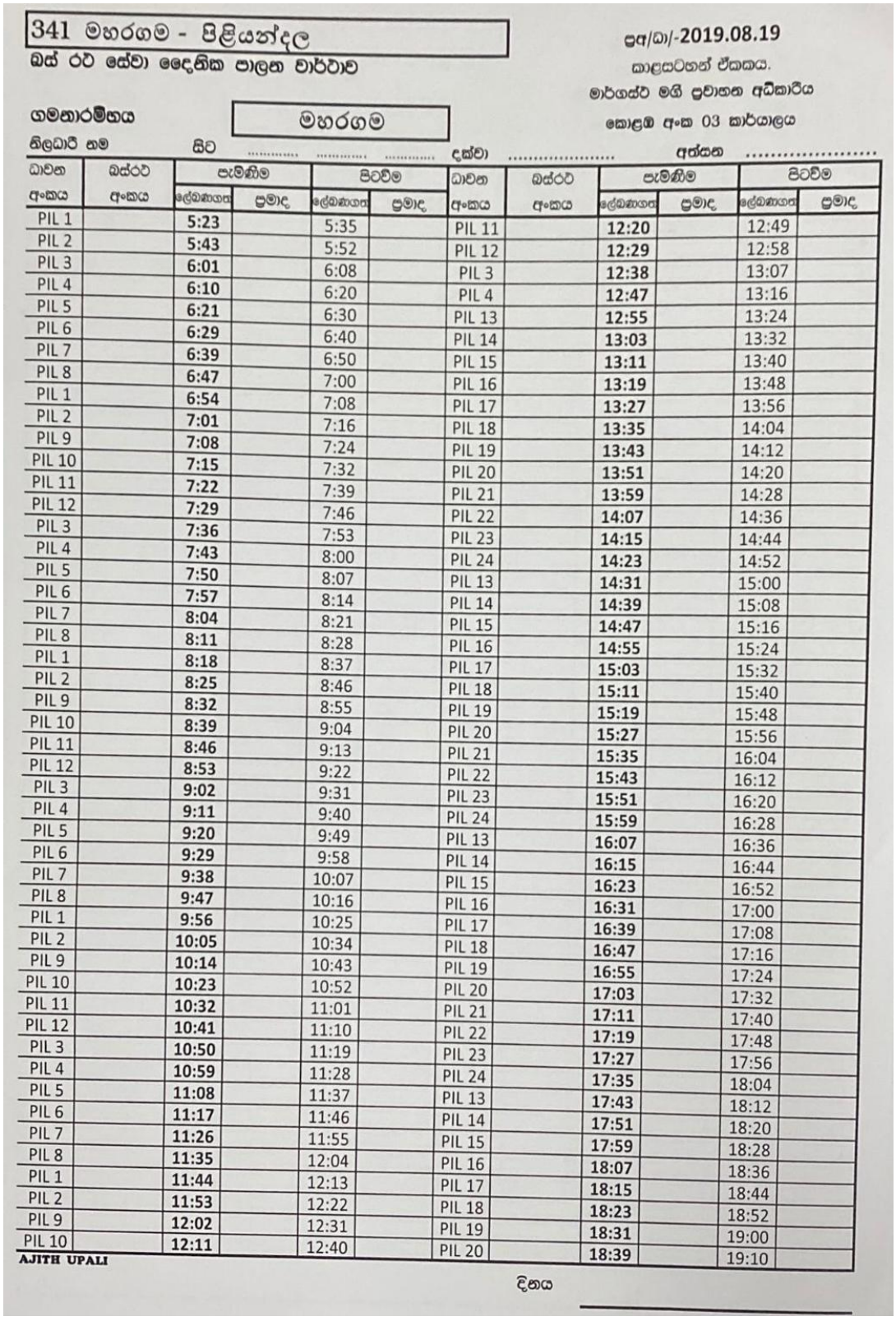
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Figure 1 - Bus Schedule



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Figure 2- Bus Schedule

The above-mentioned evidence further highlights this key issue. On this particular road (Maharagama – Piliyandala), public buses are only operated from 5AM to 7PM. Citizens who

desperately rely on public services have to look for private transportation options during off peak times, which can be highly costly. The Sri Lankan government currently has services such as Nisi Seriya that operates during off peak times. But, only a small number of buses are operated and even smaller number of routes that the buses are active, makes it a very low reliable service for passengers. These problems are affecting many stakeholders such as the government, bus owners, bus drivers and passengers. Government authorities are losing potential public revenue that could be reinvested into developing solutions for citizens' inconveniences.

In order to address these issues, we propose a GPS-based public bus route monitoring and permit revenue calculation system. A smart on-board device that combines GPS and several other sensors is installed in every bus in order to capture real life location, speed, distance travelled and driver behavior data. A central cloud platform is used to process operational data and calculate permit fees automatically using the distance-based pricing model. The system introduces an incentive mechanism that pays commissions to bus operators for running during off peak times in order to improve transportation availability. Time table making is also automated, ensuring 24-hour reliable service, and a mobile app for bus owners and drivers is developed. The app displays distance logs, off peak service commissions and permit fee estimates in order to keep the drivers and owners updated.

This research project perfectly aligns with UN sustainable development goals (SDG) 9: Innovation is done by developing new device to track the real time location. It also aligns with SDG 11: purpose is to help communities around Sri Lanka by providing 24-hour public transport service. Finally, SDG 13: The project advocates climate action with the use of environmentally friendly, technology driven public transport. The proposed project fits into research clusters such as intelligent transportation systems, data driven governance, smart cities and embedded systems.

1.2 Literature Review

A scholarly research done recently on ITS (Intelligent Transportation system) shows a significant global shift towards GPS enabled real time location monitoring and passenger information. In India, institute of science has developed a real time passenger information system which is a fundamental contribution to the public transportation field. It proposes an integrated architecture of GPS units enabled on buses, station display terminals and a centralized server that predicts arrival times of buses for passengers. The work done by them emphasizes on ETA algorithms and web/SMS passenger services in order to demonstrate how real time vehicle tracking can improve fleet efficiency and operational transparency. Their framework mainly focuses on data collections via the onboard device and using the collected data to make the travels of people more efficient by eliminating waiting time.

Generally, the existing research systems follow three main technical models. First is GPS based real time tracking model, where the device is used to transfer continuous data through cellular networks to a centralized server that compute ETAs. This approach mainly supports passenger information displays, live vehicle maps and trip planners. Second model is the offline tolerant model where the device store travel logs and upload using wi-fi to reduce communication costs Analytics driven governance model is the third. Historical data is used to evaluate driver behavior, route efficiency and traffic violations. Many systems such as OpenStreetMap use integrated digital map flatforms for route visualization and geospatial analysis.

Commercially, many passenger ISs and fleet monitoring platforms provide mature tracking services. Global ITS vendors such as Cubic Transportation systems, GMV and Siemens Mobility deliver integrated solutions like smart ticketing, vehicle location, passenger information systems and dispatch management. Transit and Moovit platforms are main global companies that aggregate live bus data to provide ETAs and service alerts. Mapping eco systems such as globally dominating Google Maps, integrate transit feeds for real time bus tracking and schedule optimization.

However, all of these systems mainly focus passenger experience, operational logistics and journey planning. Despite the sophistication of these systems, they show several limitations. Regulatory revenue models remaining mostly fixed is the key limitation. Especially in public transportation. Like in most countries including Sri Lanka, government charges fixed monthly

route permit fees or significantly rely on passenger fare taxation. None of the existing public transportation platforms compute permit revenue using a distance-based model, which created revenue leakage and inefficiencies. Plus, without a proper way to generate revenue, public transportation gaps remain unresolved. While many platforms focus on passengers, the bus owners' and drivers' needs remain unchecked. Although research prototypes have included speed and behavior monitoring, practical deployments have limited such analytics to logistics or insurance sectors. There for there is a need for proper revenue generation so the government can make citizens' transportation needs easier.

1.3 Research Gap

Existing intelligent transportation systems and solution reveal several critical gaps that remains not addressed. Most of the widely known systems focus on passenger convenience, fleet logistics and operational efficiency rather than permit fairness and government revenue optimization. Permit countries in many countries still depend on fixed monthly payments without considering the actual service frequency. This issue has created structural inefficiencies where highly profited bus owners pay the same permit fees as a low profited operator, leading to inequitable financial burdens among operators. Due to the low revenue generation, off peak public services remain a consistent issue because of existing systems' lack of attention to introduce economic incentives that align operator profit motives with public reliability needs. Therefore, current research approaches remain insufficient in several ways. Such as the GPS based tracking system proposed by Indian Institute of Science which uses vehicle telemetry to improve fleet scheduling and passenger information rather than enforcing dynamic permit models and computing regulatory revenue.

The novelty and the creativity in our proposed system mainly lie in reframing GPS based transport intelligence from a passenger- service tool into a economic governance model. Without using vehicle tracking only for ETA predictions and route visualizations, the proposed system is building distance-based revenue model. This model helps to calculate government earnings from route permits based on distance they've travelled to provide public transportation service. This innovation transforms GPS telemetry into a transparent mechanism that link road usage to public revenue. This helps eliminate flat fee distortions and ensure equity among operators. The new system also introduces an incentive aligned service framework. With this, operators are motivated to run buses during off peak times because of commission adjustments

made possible from efficient revenue generation. This mechanism combines economic nudging with operational monitoring, answering a long-standing availability problem that couldn't be resolved through passenger information systems alone.

Development of stakeholder-centered mobile app that integrates bus owners, drivers and regulators is another key innovation in this project. While all the existing systems prioritizing passengers and control rooms, the proposed mobile app gives automated payment reminders and regularly update operators on real time fee estimations and commissions.

The evidence-based justification for this project is based in both observable transport sector challenges and documented research findings. While other studies have also shown GPS enabled ITS platforms highly improve operational data accuracy and fleet visibility, the research done by professor Sugata Sanyal, professor Haresh Dagale and several others at Indian institute of Science in Bangalore, have even demonstrated the use of this technology in several areas of the country and proven its's effectiveness. Professor Saed Tarapiah of An-Najah national university of Palestine and Shadi Atalla of Lavoro Autonomo in Italy have also proven the historical mobility data sets enable violation detection and behavioral analysis. However, these researches are underutilized in public revenue systems. But evidence has shown success of usage-based pricing models in other business sectors.

2. OBJECTIVE

2.1 Main Objective

The main objective of this project is to increase the availability of public bus services during off peak hours (8 PM – 5AM) and make the transportation more reliable. During these hours of the day, the passengers face a lot of difficulties because of reduced fleet operations. Due to lower passenger count and high operating costs, private bus owners are discouraged to operate during off peak hours. This research aims to address this issue by using digital technology driven monitoring and innovative mechanisms to make operations economically more viable for operators. By aligning operator profitability with social mobility needs, the project looks to transform public bus services into a sustainable and regulated service supported by automation and data.

2.2 Specific Objectives

- Develop a GPS based tracking system to accurately monitor bus operations to improve reliability and reduce route violations.
- Design a commission payment model which pays bus owners an extra amount to encourage them to operate during off peaks times.
- Implement an automated scheduling system to ensure minimum public service frequency on key routes.
- Create a mobile application for bus owners and drivers to keep them updated about daily earning, permit fees and incentives.
- Provide responsible transport authorities with dashboards to monitor and supervise service courage to reduce route deviations and unreliable services.
- Improve the confidence and safety of passengers by developing an on-board unit with sensors that can monitor speed and aggressiveness of buses.
- Collect and analyze data in order to support long term public transportation goals and sustainability.

3. METHODOLOGY

3.1 Research Approach

This project adopts a Design Science Research (DSR) methodology, which is widely recognized as appropriate for information systems research that involves the creation and evaluation of novel IT artefacts intended to solve identified organizational or societal problems. The DSR paradigm is particularly suitable for this study because the project does not merely investigate phenomena but actively constructs a functional system comprising hardware, software, and algorithmic components that must be evaluated against real-world requirements.

3.2 Technical methods and algorithms

The system employs a range of technical methods and algorithms distributed across its four functional layers.

1. **Event based data acquisition and transmission**

Unlike conventional continuous-polling GPS trackers that transmit coordinates at fixed intervals this system implements an event-driven architecture in which the on-board IoT device transmits data packets only upon the occurrence of significant operational events such as junction entry and exit, halt detection and driver behavior.

2. **Trip segmentation and distance calculation**

Each operational trip is segmented based on the detection of trip-start and trip-end events. Within a trip, the total distance is calculated as the sum of inter-junction distances along the official route polyline. For segments where a bus deviates from the expected route, the system falls back to GPS-based Haversine cumulative distance using sampled coordinates, which is flagged as an estimated distance and excluded from the compliant distance calculation for permit fee purposes.

3. **Time Slot clarification and revenue calculation**

Based on programmable policy criteria, such as Peak, Normal and Night (Incentive), each with its own rate multiplier, each completed journey is categorized into predetermined time intervals. If a trip crosses several time-slot boundaries, the algorithm splits the distance equally depending on the timestamps recorded throughout the trip. The total trip fee is then calculated by multiplying the distance traveled inside each time slot by the applicable per-kilometer rate for that slot to guarantee accurate permit revenue computation.

4. **Off-peak commission calculation**

For trips completed during the night incentive window (22:00–05:00), the bus operator receives a government-paid incentive instead of paying a permit fee. The incentive is calculated based on the validated distance traveled during nighttime hours multiplied by the government-configured incentive rate per kilometer.

3.3 Tools and Platforms

Component	Specification	Justification
ESP32 Microcontroller	Dual-core 240 MHz, built-in Wi-Fi and Bluetooth, 520 KB SRAM	Selected as the central on-board processing unit due to its integrated Wi-Fi capability (eliminating the need for a separate communication module), low power consumption, sufficient processing power for real-time geofence calculations and IMU signal processing, The ESP32 supports the Arduino framework, enabling rapid prototyping.
NEO-6M GPS Module	u-box, 50-channel, 2.5 m CEP accuracy, 1 Hz update rate	Provides reliable NMEA sentence output at low cost. The 2.5 m horizontal accuracy is sufficient for junction-level geofencing with 50 m radii. Compatible with ESP32 via UART serial interface.
MPU6050 IMU Sensor	3-axis accelerometer, 3-axis gyroscope, I²C interface	The MPU6050 provides six degrees of freedom, communicates via I²C (requiring only 2 GPIO pins on the ESP32), and has extensive library support. Its configurable ranges allow optimization for vehicle dynamics. The built-in Digital Motion Processor (DMP) can offload some filtering computations from the ESP32.
5V Vehicle Power Supply with Voltage Regulator	LM7805 or buck converter, 12V/24V to 5V	Powers the IoT device from the bus electrical system. A

		voltage regulator ensures stable 5V supply regardless of vehicle voltage fluctuations.

Table 1- Component & justification

Platform / Tool	**Purpose**	**Justification**
Arduino IDE	Firmware development for ESP32	Used to write and upload code to the ESP32 device. Provides libraries and tools needed for GPS, sensors, and Wi-Fi communication.
Azure IoT Hub	IoT device communication and data ingestion	Allows ESP32 devices to securely send real-time data to the cloud. It manages communication between many IoT devices and the backend system.
Spring Boot	Backend REST API and business logic	Used to build the backend server that processes device data, manages APIs, and connects the database with the frontend applications.
PostgreSQL	Primary database	Stores system data such as bus routes, locations, drivers, and logs in a structured and reliable way.
React	Regulator web dashboard	Used to create an interactive web dashboard for monitoring bus locations, routes, and system data.
Ant Design (Ant D)	UI component library for web dashboard	Provides ready-made, responsive UI components to design professional and consistent dashboard interfaces efficiently.
React Native	Bus owner/driver mobile application	Allows development of a mobile application for both Android and iOS using a single codebase.
Leaflet / Map box GL JS	Map visualization	Used to display maps, bus locations, routes, and alerts visually in the dashboard and mobile app.

Table 2- Platform & Justification

Tool	Purpose	Justification
Git / GitHub	Version control and collaborative development	Used to track code changes and allow multiple developers to work on the project efficiently.
Postman	API endpoint testing and documentation	Helps test backend APIs and verify requests and responses during development.
Arduino Serial Monitor / Arduino Serial Plotter	Real-time IMU and GPS data debugging during firmware development	Used to view and debug sensor data such as GPS coordinates and motion readings from the ESP32 device.
Google Maps Roads API	Route polyline verification and road-snapping	Used to validate and align bus routes with real road paths for accurate route mapping.

Table 3- Tools & Justification

3.3 Data Collection and Validation Strategy

Data for this project is collected from multiple sources to evaluate system performance. The primary source is field testing using the prototype IoT device in personal vehicles along selected bus routes, recording GPS, IMU sensor data, and system events. Controlled driving actions such as harsh braking, rapid acceleration, and sharp turns are performed to validate driver behavior detection. Additional simulated data is generated to test special scenarios like route deviations and high event frequencies. Official route and policy data is collected from Road Passenger Transport Authority. Finally, stakeholder feedback is collected from bus drivers, owners, regulatory officers, and transport policy experts through interviews and questionnaires to assess system usability and effectiveness.

Since the system involves tracking vehicle locations and monitoring driver behavior, several ethical guidelines are followed. All participants involved in testing are informed about the purpose of the project and provide voluntary consent. Personal information is not stored; instead, vehicles and drivers are identified using anonymous labels. Data transmitted from the IoT device to the backend server is secured using encrypted communication, and database access is limited to authorized users only. The system collects only the minimum data required for its operation, with raw sensor data processed on the device before transmission. The project follows the research ethics guidelines of the Sri Lanka Institute of Information Technology, and formal approval will be obtained if external bus operators participate in future testing. Additionally, driver behavior data is used only to promote safer driving and provide constructive feedback rather than for punishment or surveillance.

The system validation follows a multi-layered approach covering hardware, algorithms, revenue calculations, system performance, and user acceptance. Hardware tests verify GPS and IMU accuracy, device reliability, and continuous operation. Algorithms for junction detection, route compliance, distance calculation, and driver behavior classification are validated against field and simulated datasets using metrics like precision, recall, F1-score, MAPE, and detection latency. Revenue and incentive computations are tested with predefined and simulated scenarios to ensure correctness. System-level performance is measured in terms of latency, API response time, concurrent device support, and data throughput. Finally, user acceptance testing with bus owners, drivers, and regulatory officers evaluates usability and system utility, targeting SUS scores above 68 and positive feedback on core functions.

4. HIGH-LEVEL SYSTEM ARCHITECTURE

The proposed system is an GPS enabled distance-based permit revenue management system that allows bus owners to pay their route permit fees according to the distance they have operated. Government authorities track the movement of the buses via the installed GPS device and the automated system calculates the distance travelled. Via the installed device, with the use of sensors such as Gyro sensor, bus speed and aggressiveness are also measured. Through the centralized server the relevant data is shared with different functions such as the mobile app in order to implement their task. The chart below explains the component interactions and the flow of data in the system.

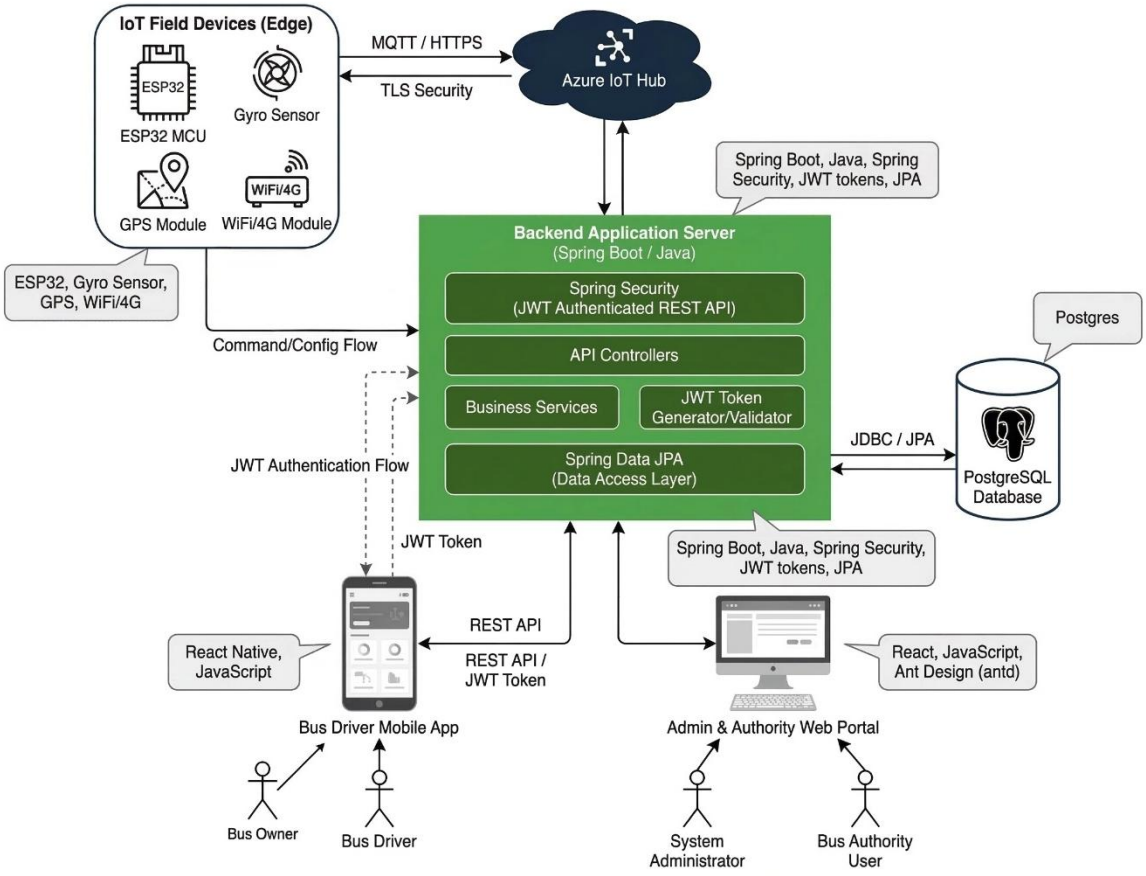


Figure 3- High-level Architecture

5. USER REQUIREMENTS

5.1 Stakeholder Identification

Main stakeholders of the proposed system include:

- Government Authorities: Wish to generate more revenue from route permits and wants to give the citizens a reliable public transportation service 24 hours.
- Bus Owners: Wish to pay a reasonable permit fee while making profit from operations.
- Bus Drivers: Wish to earn a decent living doing reasonable work
- Passengers: Wish to have a reliable bus service that operates 24 hours

5.2 Functional Requirements

FR1	User Registration and Login
Input	User credentials (Route Permit No / Username, Password)
Processing	Validate credentials, encrypt password, authenticate user, and manage user session.
Output	Successful login confirmation or error message.

Table 4- FR1

FR2	Real-Time Bus Location Tracking
Input	GPS coordinates from the bus tracking device
Processing	Receive GPS data, process location information, update system database
Output	Real-time bus location displayed on the monitoring dashboard

Table 5- FR2

FR3	Real-Time Bus Location Tracking
Input	GPS coordinates from the bus tracking device
Processing	Receive GPS data, process location information, update system database
Output	Real-time bus location displayed on the monitoring dashboard

Table 6- FR3

FR4	Trip Start and End Monitoring
Input	Trip start/end confirmation from the mobile app or GPS triggers
Processing	The system detects trip initiation and completion events and updates trip status in the database
Output	Trip status updated and displayed in the operational dashboard

Table 7- FR4

FR5	Revenue Calculation and Reporting
Input	Day and nighttime trips according to the timetable
Processing	Calculate time-based revenue to pay the government
Output	Revenue reports and financial summaries

Table 8- FR5

FR6	Notification and Alert System
Input	Route deviation alerts, system status data, and high-speed
Processing	Analyze system events and trigger notifications
Output	Alerts for operators regarding route violations or high-speed

Table 9- FR6

FR7	Notification and Alert System
Input	Route deviation alerts, system status data, and high-speed
Processing	Analyze system events and trigger notifications
Output	Alerts for operators regarding route violations or high-speed

Table 10- FR7

FR8	Admin Web Dashboard
Input	Admin login credentials
Processing	Authenticate admin and allow CRUD operations
Output	Web-based dashboard with data management features

Table 11- FR8

5.3 Non-Functional Requirements

- System Reliability: The system should operate continuously with minimal downtime to ensure uninterrupted monitoring of buses.
- Data Security: The system must protect sensitive information such as user credentials and financial data using secure authentication and encryption mechanisms.
- Performance: The system should process GPS tracking and ticketing data quickly to provide real-time updates without delays.
- Usability: The system interfaces should be easy to use for bus owners and transport authorities, with minimal training required.
- Scalability: The system should support an increasing number of buses, users, and transactions as the transportation network grows.

6. COMMERCIALIZATION PLAN

- Target market

The primary target market of the project are public transport authorities and regulators who manage private bus operations in Sri Lanka. Additional customers include private bus owners and fleet operators. The key customer persona for the project is government transport administrators who seeks better regulatory revenue, operational visibility and service reliability, and also bus owners looing for transparent permit fee calculation and profit optimization.

- Value proposition

The system helps transport authorities to equitably compute permit revenue based on actual distances traveled, as opposed to fixed fees, thus improving government revenue and transparency. The system also enhances public mobility by encouraging bus operators to operate during off-peak hours. The system helps bus owners gain real-time fee visibility, reporting, and permit services. Unlike other traditional tracking solutions, the system combines GPS, financial, service, and safety analytics in one single regulatory system.

- Revenue model

The revenue model will be a combination of a B2G and a B2B model. The government or authority will pay a platform licensing fee and a subscription fee for maintenance. The bus operators will pay a service fee for each vehicle for access to the app and connectivity to the device. Other revenue sources will be hardware sales, data analytics, customization, and long-term maintenance contracts. Premium features will be available as an add-on.

- Cost estimation

Some of the costs include hardware development costs, which cover GPS device, sensor, and embedded board costs. Other costs include software development costs, cloud or server costs, and security costs. Operational costs include the costs of maintaining the system, data storage costs, customer service costs, and periodic upgrade costs. Device installation costs, training costs, and integration with the government databases cover the deployment costs. Marketing costs, pilot costs, and compliance costs cover the costs incurred at the early stages. As the system expands, costs decrease due to economies of scale in hardware development

- Pricing strategy

The pricing strategy for the system will be a tiered model. This implies that transport authorities will be charged a scalable annual license fee, while bus operators will be charged a monthly fee for each vehicle. On the other hand, the devices will be priced using the cost-plus pricing

model or leased to the clients. Discount pricing may also apply to the pilot programs to ensure government adoption. Value-based pricing implies that the fee charged will be less than the benefits gained from increased permit revenue and improved efficiency.

- Competitive advantage

The key advantage of the system is that it focuses on financial information and service availability, as opposed to passenger information and fleet management. Distance-based permit automation, off-peak incentive schemes, and owner-regulator workflow are rarely combined in existing transport technology. It is an affordable solution that is scalable for developing regions, meeting government policy objectives. Its ability to integrate GPS analytics, digital governance, behavior incentives, and safety monitoring makes it a unique, impactful technology solution for the public sector.

- Intellectual property considerations

The possible intellectual property includes proprietary algorithms for distance-based permit fee calculation, incentive optimization models, and integration architecture. Software components such as mobile apps, analytics tools, and backend platforms can be covered under copyright law. Exclusive hardware designs and methods of integrating sensors may be eligible for design patents. Trademark law may apply to the product itself. For open-source software components such as mapping tools, licensing agreements are required. Non-disclosure agreements and government contracts will cover the operational data and designs.

7. BUDGET AND JUSTIFICATION

The estimated project cost includes software fees, hardware infrastructure, network service fees, personnel expenses and maintenance expenses required to develop the proposed System.

DESCRIPTION	AMOUNT (LKR)	TOTAL (LKR)
SOFTWARE		
React.js	6 000	
Spring Boot	10 000	
React native	10 000	
C++	7 000	
PostgreSQL	9 000	
Visual Code	5 000	
Software Cost		**47 000**
HARDWARE		
Router	6 000	
2 pc	40 000	
IoT Device	9 000	
Total Hardware Cost		**55 000**
NETWORK		
Data transfer and API usage	8 000	
Hosting and server deployment	7 000	
Total Networking Cost		**15 000**
MAINTENCE		
Client training and technical support	6 000	
Warranty and support services	10 000	
Security software and monitoring	4 000	
System maintenance agreement	5 000	
Total Maintenance Cost		**25 000**
Total Project Cost		**142 000**

Table 12- Budget Estimation

8. WORK BREAKDOWN STRUCTURE (WBS)

The Work Breakdown Structure (WBS) provides a hierarchical representation of all tasks required for the predictive warfarin dosing system. The project breaks down into its main phases, which include requirement analysis, system design, data collection model development, implementation testing, deployment, and documentation.

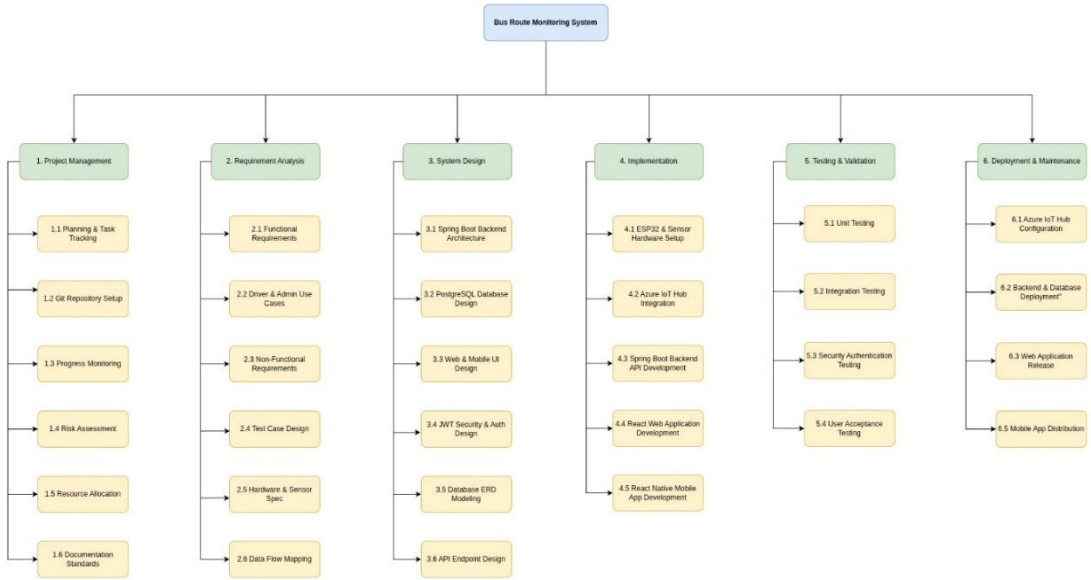


Figure 4 - WBS chart

The Work Breakdown Structure (WBS) outlines the primary tasks necessary to complete the research project.

Task ID	Task Description	Responsible Member	Timeline
T1	Literature review and problem analysis of public bus transport monitoring systems	All Members	Weeks 1-3
T2	Collection of transport operational data, route schedules, and policy information from the Road Passenger Transport Authority	All Member	Weeks 3-5
T3	Development of an IoT-based GPS event monitoring and trip data acquisition module	Member 1	Weeks 5–7
T4	Development of route compliance verification and map-matching algorithm	Member 1	Weeks 7–10
T5	Design and implementation of Policy & Time-Slot Engine for automated scheduling and service classification	Member 2	Weeks 8–11
T6	Development of an incentive and revenue calculation module based on operational distance and service periods	Member 3	Weeks 9–12
T7	Database design and implementation for route data, policy rules, and operational logs using PostgreSQL	All Members	Weeks 10–12
T8	Development of a regulator web dashboard using React for monitoring routes and policy compliance	Members 2 & 3	Weeks 11–13
T9	Development of mobile applications for bus operators to update trips and receive operational notifications	Member 4	Weeks 11–13
T10	Integration of IoT monitoring, policy engine, revenue module, and user interfaces	All Members	Weeks 14–16
T11	System testing and validation using simulated and real operational scenarios	All Members	Weeks 16–18
T12	Performance evaluation, documentation, and final system improvements	All Members	Weeks 18–20

Table 13- WBS

9. GANTT CHART

The Gantt chart shows both the research project activities schedule and their planned execution order. The document defines main project stages, which include literature review and dataset collection, model development, system implementation and testing, and final documentation.

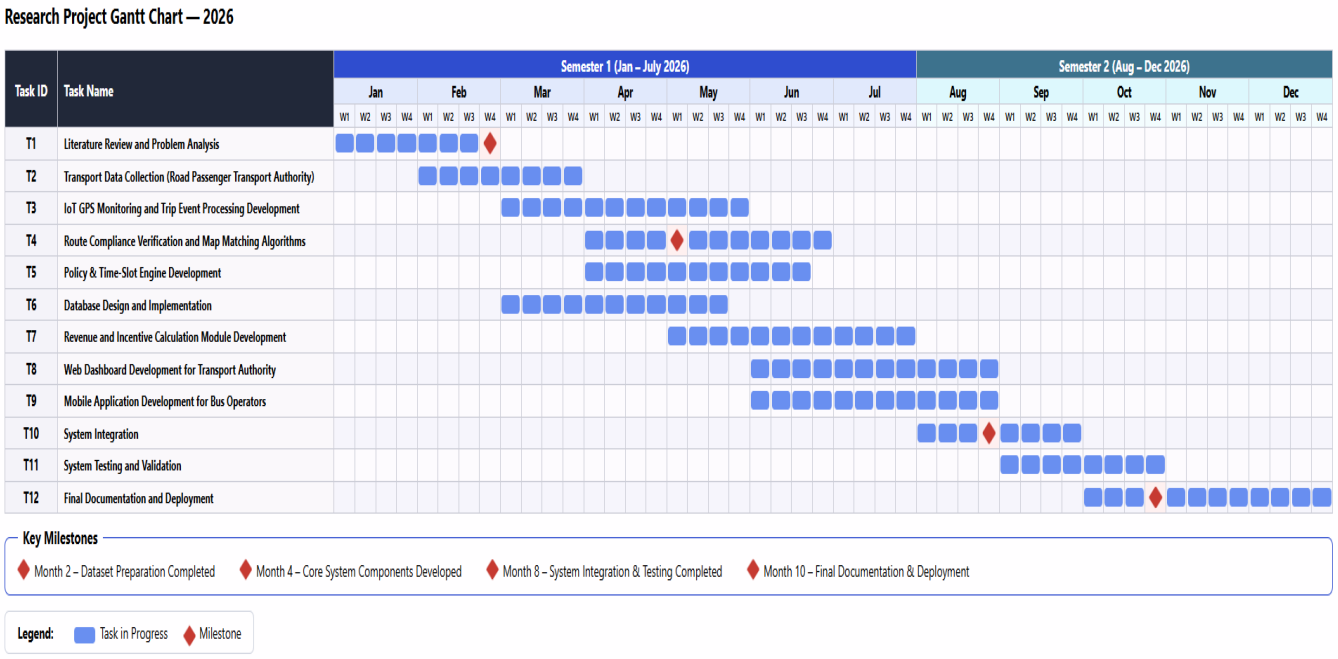


Figure 5 - Gantt Chart

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11.APPENDICES

Appendix A – Survey / Interview Instrument

The appendix contains information collected through stakeholder discussions and consultations on public transport operations. The research data and insights were primarily derived from the Road Passenger Transport Authority, which included route tracking data, scheduling information, and details about bus operation difficulties shared by the system administrator.

Appendix B - Detailed System Diagrams

The appendix presents complete technical diagrams that were utilized throughout the system development process. The system consists of three main components, which include overall system architecture, component diagrams, and process flow diagrams to demonstrate how the IoT monitoring devices work with the cloud processing layer and Policy & Time-Slot Engine in the system.

Appendix C - Preliminary UI mockups

